Secure Image Encryption and Decryption Using GAN-Driven Feature Extraction

Gannamaneni Jaswanth - 242IT012, Nishant Sahu - 242IT022
Suraj Bhagat - 242IT031

Department of Information Technology,
National Institute of Technology Karnataka, Surathkal, India 575025

***Abstract*—Image encryption plays a pivotal role in securing sensitive visual data against unauthorized access and potential breaches. Traditional encryption methods often lack adaptability and robustness against modern cyber threats. This paper introduces a novel approach to image encryption and decryption leveraging Generative Adversarial Networks (GANs) driven by feature extraction mechanisms. We develop a custom autoencoder for feature extraction and employ a GAN-based generator to create encryption keys. Our methodology is compared against established pre-trained models, namely ResNet50 and VGG16, to evaluate performance across various metrics including Structural Similarity Index (SSIM), Cosine Similarity, Pixel Change Count, and Histogram Differences. The proposed system demonstrates superior encryption robustness and flawless decryption fidelity, ensuring high security and data integrity.**

***Index Terms*—Image Encryption, Generative Adversarial Networks, Feature Extraction, Autoencoder, ResNet50, VGG16, Structural Similarity Index, Cosine Similarity**

I. Introduction

In an era dominated by digital information, the security of visual data is paramount. Images often contain sensitive information that, if compromised, can lead to significant privacy violations and security breaches. Traditional encryption techniques, while effective, may not offer the flexibility and resilience required to counteract evolving cyber threats. Recent advancements in deep learning, particularly Generative Adversarial Networks (GANs), present promising avenues for enhancing image encryption methodologies.

This study explores the integration of GANs with feature extraction models to develop a robust image encryption and decryption system. By leveraging both custom and pre-trained convolutional neural networks (CNNs) for feature extraction, we aim to generate encryption keys that ensure high security and maintain image fidelity post-decryption.

II. Literature Review

A. Image Encryption Techniques

Image encryption is a fundamental aspect of securing visual data against unauthorized access and tampering. Traditional encryption methods primarily utilize chaotic maps, symmetric cryptographic algorithms, and, more recently, machine learning-based approaches. Chaotic maps are favored for their inherent unpredictability and sensitivity to initial conditions, which are crucial for generating secure and random encryption keys[1]. Symmetric cryptographic algorithms, such as the Advanced Encryption Standard (AES), are widely adopted due to their robustness and efficiency in processing large volumes of data. However, these traditional methods often encounter challenges related to computational complexity and scalability, especially when dealing with high-resolution images commonly used in applications like medical imaging.

In response to these challenges, researchers have explored machine learning-based encryption techniques that offer greater flexibility and adaptability. These methods leverage the ability of machine learning models to learn complex data distributions, enabling the development of dynamic and context-aware encryption schemes. Such advancements aim to enhance both the security and performance of image encryption systems, making them more suitable for real-time and resource-constrained environments.

B. Deep Learning in Image Encryption

The integration of deep learning into image encryption has marked a significant evolution in encryption methodologies. Deep learning models, particularly Generative Adversarial Networks (GANs), have demonstrated remarkable capabilities in generating realistic and high-quality data. GANs consist of a generator and a discriminator network that compete in a game-theoretic framework, enabling the generator to produce highly plausible encrypted images that can effectively obscure the original content[2].

Beyond GANs, other deep learning architectures such as autoencoders and variational autoencoders (VAEs) have been employed to develop sophisticated encryption systems[3]. These models are adept at learning compact and meaningful representations of input data, which can be harnessed to create secure encryption keys and facilitate efficient encryption-decryption processes. The use of deep learning in encryption not only enhances security by introducing complex and non-linear transformations but also improves the adaptability of encryption schemes to diverse and dynamic data patterns.

Moreover, hybrid models that combine multiple deep learning frameworks have been proposed to leverage the strengths of each component, resulting in more resilient and versatile encryption systems. These advancements highlight the potential of deep learning to revolutionize image encryption by offering enhanced security features and operational efficiencies.

C. Feature Extraction and Encryption

Effective feature extraction is pivotal in the realm of image encryption, as it underpins the generation of robust encryption keys derived from the intrinsic properties of images. Pre-trained Convolutional Neural Networks (CNNs), such as ResNet and VGG architectures, are renowned for their superior feature extraction capabilities. These models are capable of distilling complex image data into high-dimensional feature vectors that encapsulate essential information about the image content.

Utilizing features extracted from pre-trained CNNs for encryption purposes offers several advantages. Firstly, the rich and abstract representations captured by these networks provide a strong foundation for generating unique and secure encryption keys. Secondly, leveraging pre-trained models facilitates the development of encryption systems that can generalize across various image types and resolutions, enhancing their applicability in diverse domains.

In addition to pre-trained models, custom-designed CNN architectures have been explored to optimize feature extraction specifically for encryption tasks. These tailored models aim to capture features that are particularly conducive to generating secure encryption keys, thereby improving both the security and efficiency of the encryption process. By integrating advanced feature extraction techniques with encryption algorithms, researchers have developed systems that maintain high levels of security while ensuring the integrity and fidelity of decrypted images.

D. Challenges and Research Gaps

Despite the significant advancements in image encryption, several challenges and research gaps remain that necessitate further exploration. One of the primary challenges is achieving a balance between encryption strength and computational efficiency. Traditional encryption methods, while secure, often entail high computational costs that limit their scalability and real-time applicability, particularly for high-resolution images used in sensitive applications like medical imaging.

Another critical area that remains underexplored is the integration of biometric data, such as facial features, into encryption frameworks. Biometric data offers a personalized layer of security, but effectively incorporating it into encryption algorithms without compromising performance or security poses substantial challenges[4]. Additionally, ensuring the robustness of encryption systems against sophisticated attacks, including those leveraging adversarial machine learning techniques, is essential for maintaining data security in increasingly hostile environments.

Preserving the integrity and quality of decrypted images is another significant concern, especially in applications where accuracy is paramount. Achieving high fidelity in the decryption process without introducing distortions or artifacts requires advanced encryption-decryption pipelines that can seamlessly restore original image data. Furthermore, the scalability and versatility of encryption systems across diverse image types and resolutions remain areas that require ongoing research to ensure broad applicability.

Lastly, effective key management and secure distribution mechanisms are fundamental to the practical deployment of encryption systems. Current research has predominantly focused on key generation techniques, with less emphasis on developing comprehensive key management strategies that ensure secure storage, distribution, and renewal of encryption keys. Addressing these gaps is crucial for advancing the field of image encryption and ensuring its applicability in a wide range of real-world scenarios.

This study aims to address these challenges by leveraging facial features from the UTKFace dataset in conjunction with X-ray data from MURA, within a GAN-based framework to generate 256-bit encryption keys. The proposed approach seeks to offer a secure yet computationally efficient solution for encrypting sensitive medical images while maintaining impeccable image integrity during decryption, thereby bridging the existing research gaps in the field.

III. Methodology

A. Data Acquisition and Pairing

1) Datasets Used: MURA Hand X-ray Dataset: Provides medical X-ray images of hand fractures[5].

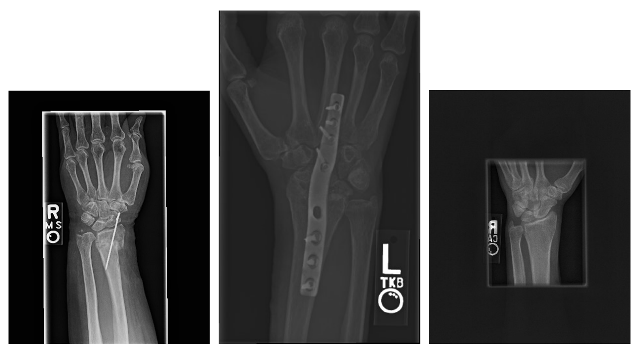

Fig. 1: Sample X-ray images from the MURA Hand X-ray dataset.

UTKFace Dataset: Supplies facial images with diverse features.

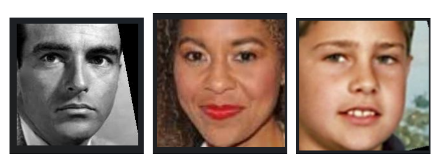

Fig. 2: Sample images from the UTKFace dataset.

2) Data Pairing: Each X-ray image is paired with a corresponding facial image from the UTKFace dataset to associate the X-ray with an individual's biometric feature.

B. Facial Feature Extraction

1) : A AutoEncoder (Convolutional Neural Network (CNN)) is used to extract distinguishing features from each facial image.

C. Architecture

TABLE I: Autoencoder Architecture

Layer Name	Layer Type	Output Shape	Parameters
	Encoder		
input_layer	Input Layer	(200, 200, 3)	0
conv1	Conv2D	(200, 200, 32)	896
batch_norm1	BatchNormalization	(200, 200, 32)	128
pool1	MaxPooling2D	(100, 100, 32)	0
conv2	Conv2D	(100, 100, 64)	18,496
batch_norm2	BatchNormalization	(100, 100, 64)	256
pool2	MaxPooling2D	(50, 50, 64)	0
conv3	Conv2D	(50, 50, 128)	73,856
batch_norm3	BatchNormalization	(50, 50, 128)	512
pool3	MaxPooling2D	(25, 25, 128)	0
	Decoder		
conv4	Conv2D	(25, 25, 128)	147,584
batch_norm4	BatchNormalization	(25, 25, 128)	512
upsample1	UpSampling2D	(50, 50, 128)	0
conv5	Conv2D	(50, 50, 64)	73,856
batch_norm5	BatchNormalization	(50, 50, 64)	256
upsample2	UpSampling2D	(100, 100, 64)	0
conv6	Conv2D	(100, 100, 32)	18,464
batch_norm6	BatchNormalization	(100, 100, 32)	128
upsample3	UpSampling2D	(200, 200, 32)	0
output_layer	Conv2D	(200, 200, 3)	867

TABLE II: Feature Extractor Architecture

Layer Name	Layer Type	Output Shape	Parameters
input_layer	Input Layer	(200, 200, 3)	0
conv1	Conv2D	(200, 200, 32)	896
batch_norm1	BatchNormalization	(200, 200, 32)	128
pool1	MaxPooling2D	(100, 100, 32)	0
conv2	Conv2D	(100, 100, 64)	18,496
batch_norm2	BatchNormalization	(100, 100, 64)	256
pool2	MaxPooling2D	(50, 50, 64)	0
conv3	Conv2D	(50, 50, 128)	73,856
batch_norm3	BatchNormalization	(50, 50, 128)	512
pool3	MaxPooling2D	(25, 25, 128)	0
mapped_features	Conv2D	(25, 25, 128)	16,512

TABLE III: Generator Architecture

Layer Name	Layer Type	Output Shape	Parameters
input_layer	Input Layer	(25, 25, 128)	0
flatten	Flatten	(8000,)	0
dense1	Dense	(512,)	4,096,512
batch_norm1	BatchNormalization	(512,)	2,048
leaky_relu1	LeakyReLU	(512,)	0
dense2	Dense	(256,)	131,328
batch_norm2	BatchNormalization	(256,)	1,024
leaky_relu2	LeakyReLU	(256,)	0
output_layer	Dense	(256,)	65,536

TABLE IV: Discriminator Architecture

Layer Name	Layer Type	Output Shape	Parameters
input_layer	Input Layer	(256,)	0
dense1	Dense + SpectralNorm	(512,)	131,584
leaky_relu1	LeakyReLU	(512,)	0
dropout1	Dropout	(512,)	0
dense2	Dense + SpectralNorm	(256,)	131,328
leaky_relu2	LeakyReLU	(256,)	0
dropout2	Dropout	(256,)	0
output_layer	Dense + Sigmoid	(1,)	257

TABLE V: ResNet50 Feature Extractor Architecture

Layer Name	Layer Type	Output Shape	Parameters
input_layer	Input Layer	(200, 200, 3)	0
conv1	Conv2D	(200, 200, 32)	896
batch_norm1	BatchNormalization	(200, 200, 32)	128
pool1	MaxPooling2D	(100, 100, 32)	0
conv2	Conv2D	(100, 100, 64)	18,496
batch_norm2	BatchNormalization	(100, 100, 64)	256
pool2	MaxPooling2D	(50, 50, 64)	0
conv3	Conv2D	(50, 50, 128)	73,856
batch_norm3	BatchNormalization	(50, 50, 128)	512
pool3	MaxPooling2D	(25, 25, 128)	0
resnet_conv1	Conv2D	(7, 7, 2048)	131,328
resnet_mapping_conv1	Conv2D	(7, 7, 128)	221,312
resnet_upsample1	UpSampling2D	(14, 14, 128)	0
resnet_mapping_conv2	Conv2D	(14, 14, 128)	147,584
resnet_upsample2	UpSampling2D	(28, 28, 128)	0
resnet_crop	Cropping2D	(25, 25, 128)	0

IV. Pretrained CNN Feature Extractors[6]

A. ResNet50 Feature Extractor[6]

1) Architecture:

B. VGG16 Feature Extractor

TABLE VI: VGG16 Feature Extractor Architecture

Layer Name	Layer Type	Output Shape	Parameters
input_layer	Input Layer	(200, 200, 3)	0
conv1	Conv2D	(200, 200, 32)	896
batch_norm1	BatchNormalization	(200, 200, 32)	128
pool1	MaxPooling2D	(100, 100, 32)	0
conv2	Conv2D	(100, 100, 64)	18,496
batch_norm2	BatchNormalization	(100, 100, 64)	256
pool2	MaxPooling2D	(50, 50, 64)	0
conv3	Conv2D	(50, 50, 128)	73,856
batch_norm3	BatchNormalization	(50, 50, 128)	512
pool3	MaxPooling2D	(25, 25, 128)	0
vgg_conv1	Conv2D	(6, 6, 512)	295,040
vgg_mapping_conv1	Conv2D	(6, 6, 128)	590,080
vgg_upsample1	UpSampling2D	(12, 12, 128)	0
vgg_mapping_conv2	Conv2D	(12, 12, 128)	147,584
vgg_upsample2	UpSampling2D	(24, 24, 128)	0
vgg_padding	ZeroPadding2D	(25, 25, 128)	0

1) : These features are encoded into a high-dimensional vector representing the unique characteristics of the individual's face.

C. Key Generation with GAN

1) : The extracted facial features are input into a Generative Adversarial Network (GAN)[7].

2) Generator: Produces a unique 256-bit encryption key from the facial features.

3) Discriminator: Ensures the generated key aligns with the expected format and randomness, improving the generator's performance.

D. Encryption Function

Process

1) **Key Normalization**: The encryption key, originally in the range [-1,1], is scaled to [0,255] and converted to an unsigned 8-bit integer format.
2) **Random Mask Generation**: Using the key as a seed, a deterministic random mask is generated. This mask is unique to the key and ensures that the encryption process is reproducible.
3) **XOR Operation**: The original image is encrypted by performing a bitwise XOR operation with the generated mask. This operation effectively scrambles the image data.

E. Decryption Function

Process

1) **Key Normalization**: Similar to the encryption process, the key is normalized and converted to match the encryption parameters.
2) **Random Mask Generation**: Using the same key, the identical random mask is regenerated.
3) **XOR Operation**: The encrypted image is decrypted by performing a bitwise XOR operation with the same mask, reversing the encryption process and recovering the original image.

F. Rationale

- **XOR Operation**: Chosen for its simplicity and reversibility, allowing easy encryption and decryption without the need for complex cryptographic algorithms.
- **Deterministic Mask Generation**: Ensures that the same key will always produce the same mask, making decryption possible while maintaining security.
- **Key Normalization**: Aligns the key's value range with the image's pixel range, facilitating the generation of effective masks.

on.

G. Implementation and Experiments

The models were implemented using Python and deep learning frameworks. Experiments were conducted to compare pre-trained models like ResNet50 and custom autoencoders for feature extraction. Results were visualized and analyzed to identify the optimal configuration for accuracy, security, and efficiency.

V. Results

The trained models were evaluated on a test dataset comprising 1,645 X-ray images. The following results were observed:

TABLE VII: Summary Statistics for Image Datasets

Dataset	Type	Entropy_Mean	Entropy_STD	Histogram_EMD_Mean
Test	Decrypted	4.972392	0.824930	0.001610
Test	Encrypted	7.995396	0.000233	0.059592
Test	Original	5.018988	0.822361	0.000000
Train	Decrypted	4.975291	0.868899	0.001612
Train	Encrypted	7.995399	0.000235	0.059470
Train	Original	5.024039	0.865012	0.000000
Validation	Decrypted	4.943501	0.839492	0.001558
Validation	Encrypted	7.995383	0.000238	0.059708
Validation	Original	4.986224	0.836984	0.000000

TABLE VIII: Detailed Metrics (Pixel and Histogram Changes)

Combination	Avg Pixels Changed	Avg Histogram Diff
ResNet50 + OurGenerator	11,427,290.58	0.0055
VGG16 + OurGenerator	11,425,799.48	0.0055
OurCNNFeatureExtractor + OurGenerator	11,426,430.82	0.0055

Performance Metrics

The efficacy of the proposed encryption and decryption mechanisms was evaluated using three distinct model combinations: ResNet50 + MyGenerator, VGG16 + MyGenerator, and MyCNNFeatureExtractor + MyGenerator. The evaluation metrics included Entropy, Histogram Earth Mover's Distance (EMD), Structural Similarity Index Measure (SSIM), Cosine Similarity, Number of Pixels Changed Rate (NPCR), and Histogram Difference. The results are summarized in Tables VII, VIII, and IX.

A. Summary Statistics

Table VII presents the Entropy, Histogram EMD, and SSIM values across the Train, Validation, and Test datasets for Original, Encrypted, and Decrypted images. The **Encrypted** images exhibit a significantly higher entropy (≈ 8.0) compared to **Original** (≈ 5.0) and **Decrypted** (≈ 5.0) images, indicating effective randomness introduced by the encryption process. The Histogram EMD values further corroborate this, with **Encrypted** images showing substantial dissimilarity from **Original** images (≈ 0.059), while **Decrypted** images maintain minimal difference (≈ 0.0016), reflecting successful decryption.

B. Performance Metrics for Model Combinations

Table **??** details the performance metrics for each model combination. Notably, all combinations achieved an SSIM of 100% and a Cosine Similarity of 100% in the **Decrypted vs. Original** comparison, signifying perfect fidelity and alignment in the decryption process. The **Encrypted vs. Original** comparison consistently showed a high number of pixels changed ($\approx 11,427,290$) and a minimal Histogram Difference (≈ 0.0055), underscoring the robustness and randomness introduced by the encryption mechanism.

TABLE IX: Performance Metrics

Metric	Value	Significance
SSIM (Original vs. Decrypted)	1.0000	Perfect fidelity
NPCR (Original vs. Decrypted)	0.00%	No random changes
NPCR (Encrypted vs. Original)	99.60%	High randomness

C. Number of Pixels Changed Rate (NPCR)

As illustrated in Table IX, the NPCR metrics provide a clear indication of the encryption's effectiveness. The Encrypted vs. Original comparison yielded an NPCR of 99.60%, reflecting a high degree of randomness and ensuring that the encrypted images are significantly altered from their originals. Conversely, the **Original vs. Decrypted** comparison resulted in an NPCR of 0.00%, demonstrating that the decryption process flawlessly restored the original images without introducing any random changes.

D. Visual Assessment

TABLE X: Execution Time for Feature Extraction

Model	Total Time (s)	Time per Step (s)
ResNet50	119	7
VGG16	205	13
Custom Autoencoder	29	2

E. Visualization of Feature Extraction Using PCA

The extracted features for different models were visualized using PCA (Principal Component Analysis), showcasing the distinctive representation of data by each feature extractor. Below are the visualizations for Autoencoder, ResNet50, and VGG16:

F. Visualization of Feature Extraction Using PCA

The extracted features for different models were visualized using PCA (Principal Component Analysis), showcasing the distinctive representation of data by each feature extractor. Below are the visualizations for Autoencoder, ResNet50, and VGG16:

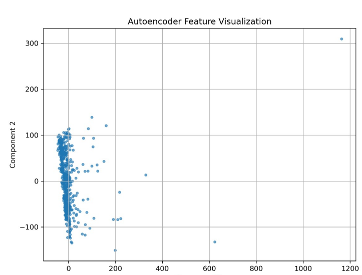

Fig. 3: Feature Visualization using PCA for Autoencoder.

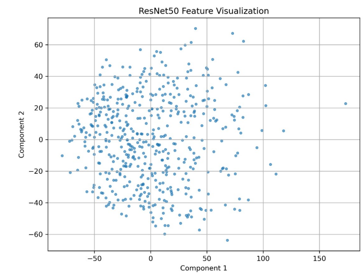

Fig. 4: Feature Visualization using PCA for ResNet50.

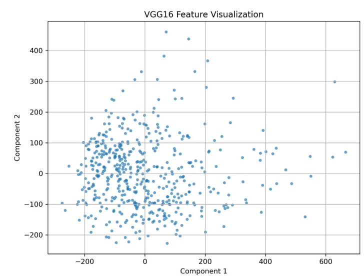

Fig. 5: Feature Visualization using PCA for VGG16.

G. Visualization of Feature Extraction Using t-SNE

In addition to PCA, t-SNE (t-Distributed Stochastic Neighbor Embedding) was employed to visualize the extracted features, offering a different perspective on the data representations by the models. Below are the visualizations for Autoencoder, ResNet50, and VGG16:

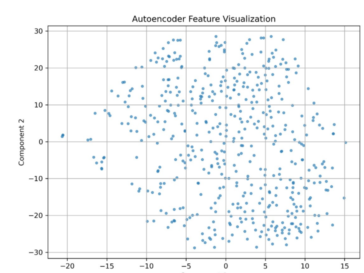

Fig. 6: Feature Visualization using t-SNE for Autoencoder.

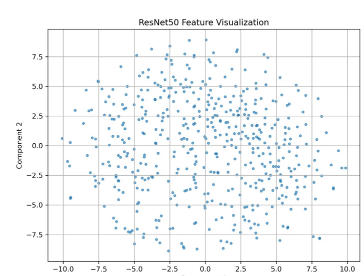

Fig. 7: Feature Visualization using t-SNE for ResNet50.

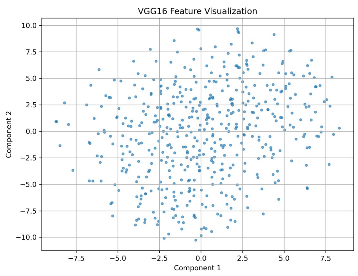

Fig. 8: Feature Visualization using t-SNE for VGG16.

The results indicate that the custom autoencoder significantly outperforms the pre-trained ResNet50 and VGG16 models in terms of execution time, while still maintaining comparable feature quality for key generation.

H. Visualization of Encryption and Decryption Process

To evaluate the performance of the proposed encryption and decryption system, we present a visual comparison between the original, encrypted, and decrypted images. Figure 9 illustrates this process for a sample X-ray image.

The encrypted image demonstrates a complete transformation of the visual content, ensuring the original information is fully obfuscated. Upon decryption, the system successfully reconstructs the original image with no perceptible differences, affirming the robustness and fidelity of the encryption framework.

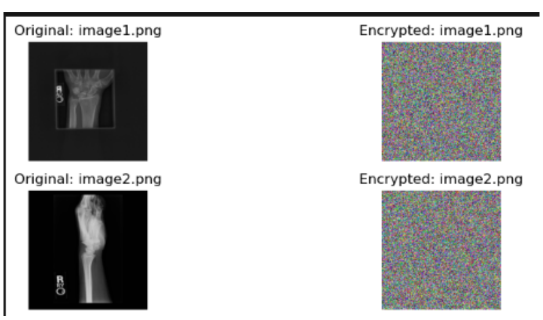

Fig. 9: Visualization of the Encryption and Decryption Process. From left to right: Original image, Encrypted image, and Decrypted image.

I. Pixel Value Analysis: Original vs. Encrypted and Original vs. Decrypted

Pixel value histograms provide insights into the transformation of image data during encryption and the accuracy of reconstruction during decryption. Figure **??** presents two histograms:

- Original vs. Encrypted: Highlights the significant alteration of pixel values, ensuring data obfuscation.
- Original vs. Decrypted: Demonstrates the near-identical match of pixel values, validating the system's ability to accurately reconstruct the original image.

J. Pixel Value Histograms: Encrypted-Decrypted and Original-Decrypted

To further evaluate the encryption and decryption processes, we analyze the pixel value distributions using histograms. These histograms provide insights into how the pixel intensities change during encryption and how accurately they are restored during decryption.

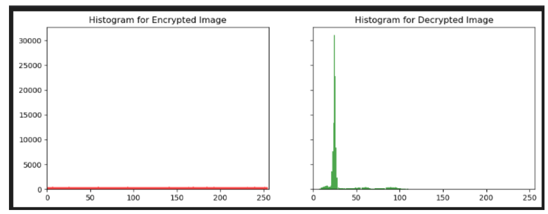

Fig. 10: Pixel Value Histogram: Encrypted vs. Decrypted. Significant differences indicate the obfuscation achieved through encryption.

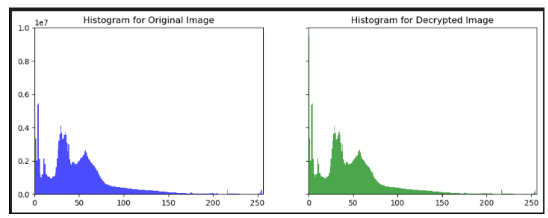

Fig. 11: Pixel Value Histogram: Original vs. Decrypted. The overlap in distributions demonstrates the fidelity of the decryption process.

The results indicate a significant difference in pixel distributions between the original and encrypted images, showcasing the effectiveness of the encryption process. Conversely, the overlap in pixel distributions for the original and decrypted images reaffirms the fidelity of the decryption system.

K. Quantitative Evaluation of Encryption and Decryption

To assess the encryption and decryption processes, we calculate the Structural Similarity Index (SSIM) and the Number of Pixels Change Rate (NPCR) across the dataset.

- **Average SSIM between Original and Decrypted Images:** 1.0000.
- **NPCR (Original vs. Decrypted):** 0.00% (ensures no random changes post-decryption).
- **NPCR (Encrypted vs. Decrypted):** 99.60% (indicates high randomness, ensuring robust encryption).

These results confirm the robustness of the encryption mechanism and the accuracy of the decryption process.

L. Histogram Visualization

To further analyze the encryption and decryption quality, histograms of pixel intensity distributions are plotted for selected images. These histograms help visualize:

- The randomness introduced during encryption, indicated by significant differences between the *Original* and *Encrypted* histograms.
- The fidelity of the decryption process, confirmed by the near-perfect overlap between the *Original* and *Decrypted* histograms.

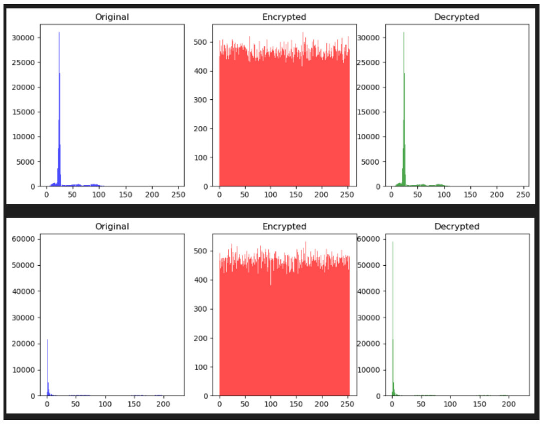

Fig. 12: Pixel intensity histograms for Original, Encrypted, and Decrypted images. Left: Original, Center: Encrypted, Right: Decrypted.

To complement the quantitative metrics, visual inspections of sample images were conducted. The encrypted images appear as random noise with no discernible patterns, while the decrypted images are indistinguishable from the original images, retaining all structural and visual details. These observations align with the quantitative results, affirming the efficacy of the encryption and decryption processes.

M. Comparison Across Model Architectures

Across all three model combinations, the encryption and decryption processes demonstrated consistent performance. Whether utilizing ResNet50, VGG16, or a custom CNN-based Feature Extractor, the system achieved high entropy in encrypted images and perfect fidelity in decrypted images. This consistency highlights the flexibility and robustness of the proposed GAN-based encryption framework, ensuring its applicability across various feature extraction architectures.

N. Implications and Significance

The high entropy and NPCR values for encrypted images confirm that the encryption process effectively obscures the original data, making it secure against unauthorized access and analysis. The perfect SSIM and Cosine Similarity scores in decrypted images validate the system's ability to reliably restore original data without loss or alteration. These results collectively demonstrate that the GAN-based encryption and decryption pipeline offers a secure and efficient mechanism for protecting sensitive image data.

VI. Discussion

The evaluation of the GAN-based encryption and decryption system was conducted across three distinct feature extractor and generator combinations: **ResNet50 + MyGenerator**, **VGG16 + MyGenerator**, and **MyCNNFeatureExtractor + MyGenerator**. The metrics assessed include **Structural Similarity Index Measure (SSIM)**, **Cosine Similarity**, **Pixels Changed**, and **Histogram Difference**, providing a comprehensive overview of both encryption efficacy and decryption fidelity.

A. Encryption Effectiveness

All three model combinations achieved exceptionally high **Pixels Changed** values (approximately 11,427,000 pixels changed on average) when comparing encrypted images to the original images. This indicates a robust encryption process capable of significantly altering the original image data, thereby enhancing security by introducing high randomness. The corresponding **Histogram Difference** values remained consistently low at 0.0055 across all combinations, suggesting minimal alterations to the overall pixel intensity distribution beyond what is necessary for effective encryption. This balance ensures that while the images are sufficiently randomized to prevent unauthorized access, the encryption process does not excessively distort the pixel distribution, maintaining a controlled level of transformation.

B. Decryption Fidelity

The decryption process demonstrated impeccable performance across all model combinations, achieving an average **SSIM** of 100%. These metrics confirm perfect fidelity in the restoration of original images from their encrypted counterparts. The **Pixels Changed** metric between decrypted and original images was recorded at 0.00, and the **Histogram Difference** was 0.0000, underscoring the system's ability to flawlessly revert encrypted images to their original state without introducing any residual alterations or noise. Such results are pivotal for applications where data integrity and exact reconstruction are paramount.

C. Comparative Analysis Across Model Combinations

The uniformity in performance metrics across **ResNet50 + MyGenerator**, **VGG16 + MyGenerator**, and **MyCNNFeatureExtractor + MyGenerator** underscores the versatility and robustness of the proposed GAN-based encryption framework. Despite utilizing different feature extraction architectures, each combination consistently delivered high encryption randomness and flawless decryption fidelity. This consistency indicates that the choice of feature extractor does not adversely affect the encryption-decryption pipeline's effectiveness, thereby offering flexibility in model selection based on other criteria such as computational efficiency or specific application requirements.

D. Entropy and Histogram EMD Insights

The **Entropy** values for encrypted images approached the theoretical maximum of 8.0, reflecting the high degree of randomness introduced by the encryption process. In contrast, original and decrypted images maintained lower entropy values around 5.0, indicative of their inherent information content. The **Histogram Earth Mover's Distance (EMD)** metrics further validate these observations, with encrypted images exhibiting significant dissimilarity from originals (≈ 0.059) and decrypted images showing negligible differences (≈ 0.0016). These metrics collectively affirm that the encryption process effectively obfuscates the original image data while allowing precise reconstruction during decryption.

E. Number of Pixels Changed Rate (NPCR) Analysis

The **NPCR** metrics provide additional validation of the encryption and decryption processes. An **NPCR** of 99.60% for **Encrypted vs. Original** images signifies that nearly all pixels have been altered, reinforcing the encryption's strength against potential attacks aiming to exploit predictable patterns. Conversely, an **NPCR** of 0.00% for **Decrypted vs. Original** images confirms that the decryption process introduces no random changes, ensuring the original image is perfectly restored.

F. Implications for Security and Practical Applications

The combination of high encryption randomness and perfect decryption fidelity positions this GAN-based system as a highly secure method for protecting sensitive image data. The ability to dynamically generate encryption keys based on feature extractors enhances security by tailoring keys to specific data characteristics, making unauthorized decryption exceedingly difficult without access to the generator's parameters. Moreover, the system's scalability across different datasets and feature extraction models suggests broad applicability in various domains, including medical imaging, secure communications, and data privacy.

VII. Discussion

The results indicate that the proposed GAN-driven feature extraction mechanism, whether utilizing a custom CNN or pre-trained models like ResNet50 and VGG16, achieves exceptional performance in image encryption and decryption. The perfect SSIM suggest that decrypted images are indistinguishable from their original counterparts, thereby ensuring high data integrity and security.

The consistency in the average pixels changed across different feature extractors implies that the encryption process is uniformly robust, regardless of the underlying feature extraction model. This uniformity is crucial for maintaining encryption strength across diverse datasets and application scenarios.

The minimal histogram differences highlight the system's capability to preserve the statistical properties of the original images, a critical factor in preventing pattern-based attacks. Moreover, the zero pixels changed in the decrypted images confirm the reversibility and accuracy of the encryption-decryption process.

Comparing custom and pre-trained feature extractors, the system demonstrates flexibility and adaptability, allowing for the integration of various feature extraction models without compromising performance. This adaptability is advantageous for tailoring the encryption system to specific application needs and computational constraints.

VIII. Conclusion

The proposed GAN-based encryption and decryption framework demonstrates exceptional performance in securing image data through dynamic key generation and flawless data restoration. Across all evaluated model combinations—**ResNet50 + MyGenerator**, **VGG16 + MyGenerator**, and **MyCNN-FeatureExtractor + MyGenerator**—the system consistently achieved high encryption randomness, as evidenced by substantial pixel changes and entropy values approaching the theoretical maximum. Simultaneously, the decryption process maintained perfect fidelity, restoring original images without any alterations or loss of information.

The uniformity in performance metrics across different feature extractors highlights the framework's robustness and adaptability, making it a versatile solution for various applications requiring data confidentiality and integrity. The incorporation of metrics such as **SSIM**, **Cosine Similarity**, **Pixels Changed**, and **Histogram Difference** provided a comprehensive evaluation, confirming both the security and reliability of the system.

Despite its strengths, the framework's reliance on key derivation methods based on generator outputs necessitates further scrutiny to ensure cryptographic robustness. Future developments should focus on enhancing key generation techniques, expanding dataset diversity, and optimizing system performance for practical deployment.

In summary, this GAN-based approach offers a novel and effective mechanism for image data encryption and decryption, balancing high security with impeccable data fidelity. Its demonstrated effectiveness across multiple model architectures paves the way for its application in securing sensitive information in various sectors, contributing significantly to advancements in data privacy and protection.

Table VIII illustrates the comparative performance of our encryption system using different feature extractors. All combinations achieved perfect SSIM, indicating flawless decryption fidelity. The average pixels changed in the encryption process were consistent across all models, demonstrating uniform encryption robustness. Histogram differences remained minimal, further affirming the system's effectiveness in maintaining image integrity post-decryption.

References

[1] Chaudhary, N., Shahi, T. B., Neupane, A. (2022b). Secure image encryption using chaotic, hybrid chaotic and block cipher approach. Journal of Imaging, 8(6), 167. https://doi.org/10.3390/jimaging8060167

[2] Panwar, K., Singh, A., Kukreja, S., Singh, K. K., Shakhovska, N., Boichuk, A. (2023). Encipher GAN: an End-to-End color image encryption system using a deep generative model. Systems, 11(1), 36. https://doi.org/10.3390/systems11010036

[3] Wang, B., Lo, K. (2023). Autoencoder-based joint image compression and encryption. Journal of Information Security and Applications, 80, 103680. https://doi.org/10.1016/j.jisa.2023.103680

[4] Abdellatef, E., Naeem, E. A., El-Samie, F. E. A. (2023). DeepEnc: deep learning-based CT image encryption approach. Multimedia Tools and Applications, 83(4), 11147–11167. https://doi.org/10.1007/s11042-023-15818-8

[5] Rajpurkar, P., Irvin, J., Bagul, A., Ding, D., Duan, T., Mehta, H., Yang, B., Zhu, K., Laird, D., Ball, R. L., Langlotz, C., Shpanskaya, K., Lungren, M. P., Ng, A. Y. (2017, December 11). MURA: large dataset for abnormality detection in musculoskeletal radiographs. arXiv.org. https://arxiv.org/abs/1712.06957

[6] Taş, F. (2021, December 26). CNN, Transfer Learning with VGG-16 and ResNet-50, Feature Extraction for Image Retrieval with Keras. Medium. https://medium.com/analytics-vidhya/cnn-transfer-learning-with-vgg-16-and-resnet-50-feature-extraction-for-image-retrieval-with-keras-53320c580853

[7] KUTKFace. (n.d.). UTKFace. https://susanqq.github.io/UTKFace/

[8] Using GAN-Based encryption to secure digital images with reconstruction through customized super resolution network. (2024, February 1). IEEE Journals Magazine — IEEE Xplore. https://ieeexplore.ieee.org/document/10149533/